

Technical Documentation Report

NexusDrive – Distributed File Storage System

1. Introduction

1.1 Purpose

This document provides a comprehensive technical overview of the **NexusDrive system**, combining both **system architecture** and **UI prototype design**. It explains how the platform enables secure file storage, management, and sharing through an intuitive user interface.

1.2 Scope

The documentation covers:

- System architecture and technologies
- Core functional modules
- Login interface (authentication layer)
- Dashboard interface (file management workspace)

2. System Overview

NexusDrive is a **cloud-based distributed file storage system** that allows users to:

- Upload and store files securely
- Organize files into directories
- Share files using secure links or permissions
- Access files efficiently from a centralized dashboard

3. System Architecture

3.1 Architecture Type

- Client-Server Model
- Cloud-based Distributed System
- RESTful API Architecture

3.2 Core Components

Frontend

- Built with React.js
- Handles UI rendering and user interactions

Backend

- Node.js / Python (Django/Flask)
- Processes requests and manages business logic

Database

- PostgreSQL / MongoDB
- Stores user data, file metadata, and permissions

Cloud Storage

- AWS S3 / Google Cloud
- Handles distributed file storage

Authentication

- JWT / OAuth 2.0
- Ensures secure user access

4. Core Modules

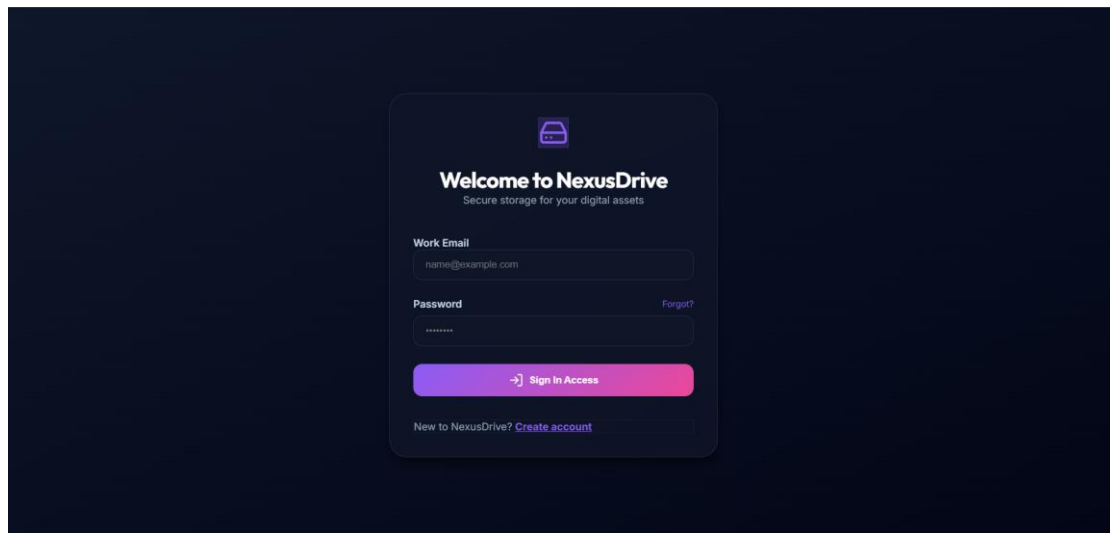
- **User Management:** Authentication, roles, profiles
- **File Storage:** Upload and distributed storage
- **File Management:** Organize, edit, delete files
- **File Sharing:** Secure links and permissions
- **Search Module:** File retrieval
- **Admin Module:** Monitoring and control

5. UI Prototype Documentation

5.1 Login Interface (Authentication Layer)

Overview

The login screen acts as the **entry point** to the system, providing secure authentication and user onboarding.



UI Components

- Centered login card with dark theme
- Email and password input fields
- “Forgot Password” option
- Gradient “Sign In” button
- “Create Account” link for new users

Functionality

- Validates user credentials via backend APIs
- On success → redirects to dashboard
- On failure → displays error messages

Security Features

- Password masking
- Secure HTTPS communication
- JWT/OAuth-based authentication

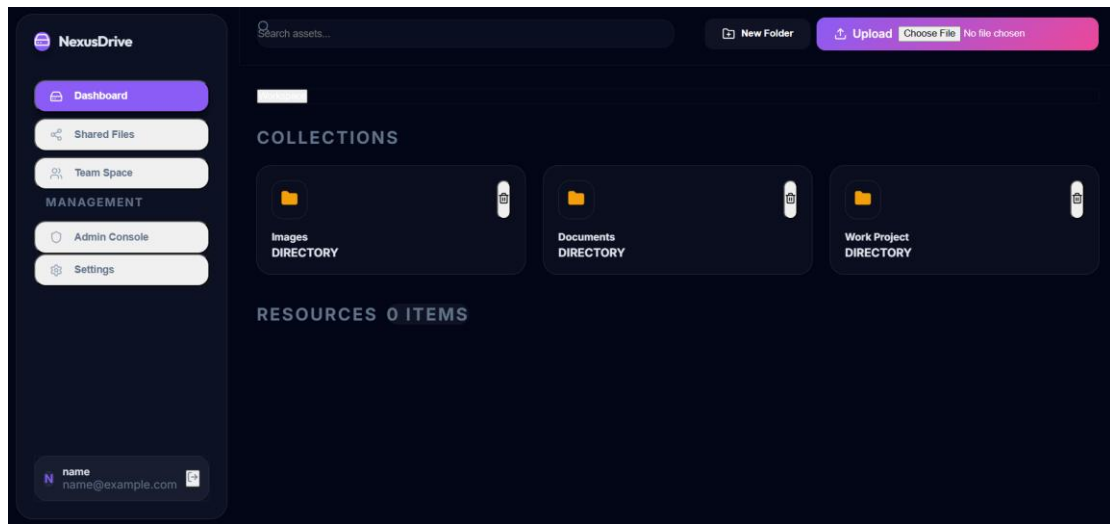
Design Highlights

- Minimal and focused layout
- High contrast for readability
- Smooth onboarding experience

5.2 Dashboard Interface (Main Workspace)

Overview

The dashboard is the **central workspace** where users manage files and navigate the system.



Layout Structure

Sidebar Navigation

- Dashboard (active view)
- Shared Files
- Team Space
- Admin Console
- Settings
- Includes user profile section at the bottom

Main Content Area

Workspace Header

- Displays current section context

Collections Section

- Displays directories as cards:
 - Images Directory
 - Documents Directory
 - Work Project Directory
- Each directory supports file organization and actions

Resources Section

- Displays file contents within selected directories
- Shows placeholder when empty (“0 Items”)

Functional Behavior

- Users navigate via sidebar
- Click folders to access files
- Perform actions like upload, delete, and organize files

Design Principles

- Dark theme for modern UI
- Card-based layout for clarity
- Clean and minimal interface
- Responsive and scalable design

6. User Workflow

1. User accesses login page
2. Enters credentials and authenticates
3. Redirected to dashboard
4. Uploads and organizes files
5. Shares files securely
6. Searches and retrieves files

7. Security Considerations

- JWT-based authentication
- Role-based access control
- Secure API communication (HTTPS)
- Input validation and sanitization

8. Deployment Strategy

- Cloud deployment (AWS / Azure / GCP)
- CI/CD pipeline with Git
- Scalable infrastructure with load balancing

9. Advantages of the System

- Scalable distributed storage

- Secure file access and sharing
- Intuitive and modern UI
- Efficient file organization
- Ready for enterprise-level expansion

10. Future Enhancements

- AI-based file organization
- Real-time collaboration
- End-to-end encryption
- Mobile application support

11. Conclusion

The NexusDrive system integrates a **robust backend architecture** with a **modern and user-friendly interface**. The login and dashboard designs ensure secure access and efficient file management, making the platform scalable, reliable, and ready for real-world deployment.